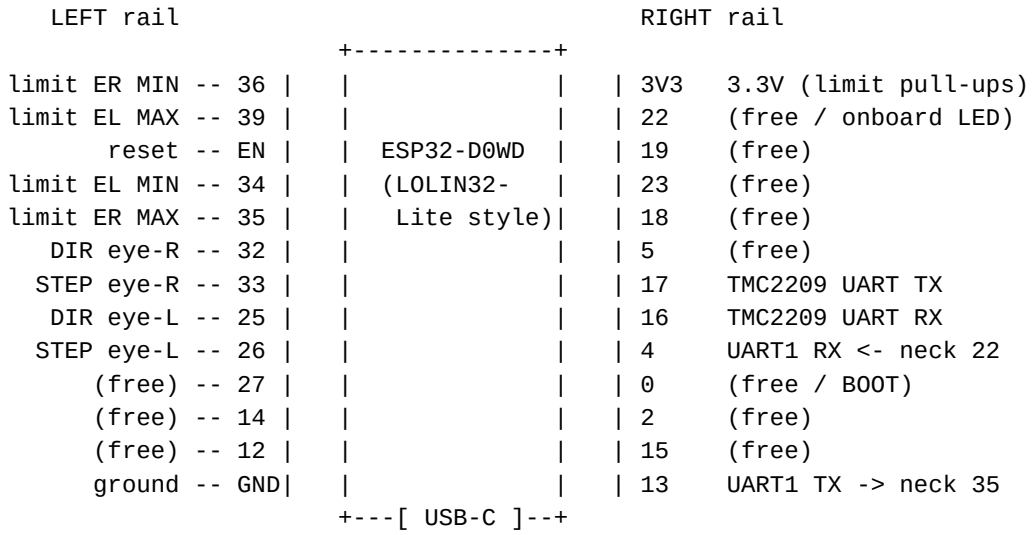
j4_stepper_eyes -- pin diagram

ESP32-D0WD (LOLIN32-Lite style), component side up, USB-C at the BOTTOM



Both TMC2209 share the UART bus (17/16); addresses 0b00 (eye-L) / 0b01 (eye-R).
Address straps: eye-L MS1 -> GND, MS2 -> GND ; eye-R MS1 -> VIO, MS2 -> GND.
GPIO 16 (RX) sits on the PDN_UART line; GPIO 17 (TX) reaches it through 1k.
Limits: one NC switch per endpoint, each with a 10k pull-up to 3.3V
(input-only pins have no internal pull-up).

GPIO	Function
26 / 25	eye-left STEP / DIR
33 / 32	eye-right STEP / DIR
17 / 16	TMC2209 UART TX / RX (shared PDN_UART bus)
4 / 13	Upstream UART1 RX / TX (to neck board)
34	eye-left MIN limit (input-only, external 10k pull-up)
39	eye-left MAX limit (input-only, external 10k pull-up)
36	eye-right MIN limit (input-only, external 10k pull-up)
35	eye-right MAX limit (input-only, external 10k pull-up)